DATABASE MANAGEMENT SYSTEM

CAPSTONE PROJECT

Aryan_2401010005
Btech cse

✈ Flight Reservation System

✈ Project Description — Flight Reservation System

This project focuses on designing and implementing a relational database system for a Flight Reservation System using MySQL. The system models core entities involved in airline operations, including airports, airlines, routes, flights, passengers, bookings, seats, payments, loyalty programs, and fare classes. Each table is designed with appropriate primary keys, foreign keys, and constraints to maintain data integrity and establish clear relationships between the components of the reservation system.

SQL queries are implemented to handle essential operations such as managing seat availability and booking status, processing flight cancellations, handling overbooking situations, prioritizing passenger upgrades based on fare class or loyalty tier, and generating payment records. These queries help minimize scheduling conflicts, automate upgrade decisions, and ensure accurate revenue tracking across flights.

The project demonstrates how a well-structured relational database can power a production-grade flight reservation platform — enabling real-time seat management, loyalty program integration, and seamless passenger experience from booking to boarding.

✈ Key Objectives

- Design a schema to store flight information including routes, schedules, and seat capacities
- Implement queries to handle seat availability, booking status, and flight cancellations
- Handle overbooking situations and prioritize passenger upgrades based on loyalty programs or fare classes
- Track payment records, refund statuses, and generate billing summaries per booking

- Manage passenger profiles and link them to loyalty reward points and tier levels
- Support multi-leg journey bookings with layover information across routes

✈ Project Information

Project Title	Flight Reservation System
Technology	MySQL (Relational Database)
Domain	Airlines / Travel Management
Key Entities	Airports, Airlines, Flights, Routes, Passengers, Bookings, Seats, Payments, Loyalty Programs, Fare Classes
Core Features	Seat booking, Overbooking management, Fare class upgrades, Loyalty points, Cancellations & Refunds

✈ Database Schema Overview

Table	Key Columns
Airports	`airport_id (PK), airport_code, name, city, country`
Airlines	`airline_id (PK), airline_name, iata_code, country`
Routes	`route_id (PK), origin_airport_id (FK), destination_airport_id (FK), distance_km`
Flights	`flight_id (PK), airline_id (FK), route_id (FK), departure_time, arrival_time, status, total_seats`
Passengers	`passenger_id (PK), full_name, email, phone, passport_no, nationality`
Fare_Classes	`fare_class_id (PK), class_name (Economy/Business/First), base_multiplier`
Seats	`seat_id (PK), flight_id (FK), seat_number, fare_class_id (FK), status (Available/Booked/Reserved)`
Bookings	`booking_id (PK), passenger_id (FK), seat_id (FK), flight_id (FK), booking_date, booking_status`
Payments	`payment_id (PK), booking_id (FK), amount, payment_status, payment_method, payment_date`
Loyalty_Programs	`loyalty_id (PK), passenger_id (FK), tier (Bronze/Silver/Gold), points_balance`
Cancellations	`cancellation_id (PK), booking_id (FK), reason, refund_status, cancelled_at`

✈ CODE

✈ CREATE DATABASE

```
CREATE DATABASE FlightReservationDB;
USE FlightReservationDB;
```

✈ CREATE TABLE

✈ Airports Table

```
CREATE TABLE Airports (
    airport_id    INT AUTO_INCREMENT PRIMARY KEY,
    airport_code  VARCHAR(10) NOT NULL UNIQUE,
    name          VARCHAR(100) NOT NULL,
    city          VARCHAR(60) NOT NULL,
    country       VARCHAR(60) NOT NULL
);
```

✈ Airlines Table

```
CREATE TABLE Airlines (
    airline_id    INT AUTO_INCREMENT PRIMARY KEY,
    airline_name  VARCHAR(100) NOT NULL,
    iata_code     VARCHAR(5) NOT NULL UNIQUE,
    country       VARCHAR(60)
);
```

✈ Routes Table

```
CREATE TABLE Routes (
    route_id               INT AUTO_INCREMENT PRIMARY KEY,
    origin_airport_id      INT NOT NULL,
    destination_airport_id INT NOT NULL,
    distance_km            DECIMAL(8,2),
    FOREIGN KEY (origin_airport_id)      REFERENCES Airports(airport_id),
    FOREIGN KEY (destination_airport_id) REFERENCES Airports(airport_id)
);
```

✈ Flights Table

```
CREATE TABLE Flights (
    flight_id      INT AUTO_INCREMENT PRIMARY KEY,
    airline_id     INT NOT NULL,
    route_id       INT NOT NULL,
    flight_number  VARCHAR(15) NOT NULL UNIQUE,
    departure_time DATETIME NOT NULL,
    arrival_time   DATETIME NOT NULL,
    total_seats    INT NOT NULL DEFAULT 180,
    status         ENUM('Scheduled','Delayed','Cancelled','Completed') DEFAULT
'Scheduled',
    FOREIGN KEY (airline_id) REFERENCES Airlines(airline_id),
    FOREIGN KEY (route_id)   REFERENCES Routes(route_id)
);
```

✈ Passengers Table

```
CREATE TABLE Passengers (
    passenger_id  INT AUTO_INCREMENT PRIMARY KEY,
    full_name     VARCHAR(100) NOT NULL,
    email         VARCHAR(100) UNIQUE NOT NULL,
    phone         VARCHAR(20),
    passport_no   VARCHAR(20) UNIQUE,
    nationality   VARCHAR(60),
    date_of_birth DATE
);
```

✈ Fare_Classes Table

```
CREATE TABLE Fare_Classes (
    fare_class_id   INT AUTO_INCREMENT PRIMARY KEY,
    class_name      ENUM('Economy','Business','First') NOT NULL,
    base_multiplier DECIMAL(4,2) DEFAULT 1.00
);
```

✈ Seats Table

```
CREATE TABLE Seats (
    seat_id       INT AUTO_INCREMENT PRIMARY KEY,
    flight_id     INT NOT NULL,
    seat_number   VARCHAR(5) NOT NULL,
    fare_class_id INT NOT NULL,
    status        ENUM('Available','Booked','Reserved') DEFAULT 'Available',
    FOREIGN KEY (flight_id)     REFERENCES Flights(flight_id),
```

```
    FOREIGN KEY (fare_class_id) REFERENCES Fare_Classes(fare_class_id)
);
```

✈ Bookings Table

```
CREATE TABLE Bookings (
    booking_id     INT AUTO_INCREMENT PRIMARY KEY,
    passenger_id   INT NOT NULL,
    seat_id        INT NOT NULL,
    flight_id      INT NOT NULL,
    booking_date   DATETIME DEFAULT CURRENT_TIMESTAMP,
    booking_status ENUM('Confirmed','Pending','Cancelled','Waitlisted') DEFAULT
'Pending',
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id),
    FOREIGN KEY (seat_id)      REFERENCES Seats(seat_id),
    FOREIGN KEY (flight_id)    REFERENCES Flights(flight_id)
);
```

✈ Payments Table

```
CREATE TABLE Payments (
    payment_id     INT AUTO_INCREMENT PRIMARY KEY,
    booking_id     INT NOT NULL,
    amount         DECIMAL(10,2) NOT NULL,
    payment_method ENUM('CreditCard','DebitCard','NetBanking','UPI','Wallet') NOT
NULL,
    payment_status ENUM('Paid','Pending','Refunded','Failed') DEFAULT 'Pending',
    payment_date   DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)
);
```

✈ Loyalty_Programs Table

```
CREATE TABLE Loyalty_Programs (
    loyalty_id      INT AUTO_INCREMENT PRIMARY KEY,
    passenger_id    INT NOT NULL UNIQUE,
    tier            ENUM('Bronze','Silver','Gold','Platinum') DEFAULT 'Bronze',
    points_balance  INT DEFAULT 0,
    enrolled_date   DATE DEFAULT (CURDATE()),
    FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id)
);
```

✈ Cancellations Table

```
CREATE TABLE Cancellations (
```

```
    cancellation_id INT AUTO_INCREMENT PRIMARY KEY,
    booking_id      INT NOT NULL,
    reason          TEXT,
    refund_status   ENUM('Pending','Processed','Denied') DEFAULT 'Pending',
    cancelled_at    DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id)
);
```

✈ INSERTION OF SAMPLE DATA

✈ Insert Airports

```
INSERT INTO Airports (airport_code, name, city, country) VALUES
('DEL', 'Indira Gandhi International', 'New Delhi', 'India'),
('BOM', 'Chhatrapati Shivaji Maharaj International', 'Mumbai', 'India'),
('DXB', 'Dubai International', 'Dubai', 'UAE'),
('LHR', 'Heathrow Airport', 'London', 'UK'),
('JFK', 'John F. Kennedy International', 'New York', 'USA');
```

✈ Insert Airlines

```
INSERT INTO Airlines (airline_name, iata_code, country) VALUES
('Air India', 'AI', 'India'),
('IndiGo', '6E', 'India'),
('Emirates', 'EK', 'UAE'),
('British Airways', 'BA', 'UK');
```

✈ Insert Fare Classes

```
INSERT INTO Fare_Classes (class_name, base_multiplier) VALUES
('Economy', 1.00),
('Business', 2.50),
('First', 4.00);
```

✈ Insert Passengers

```
INSERT INTO Passengers (full_name, email, phone, passport_no, nationality) VALUES
('Aarav Sharma', 'aarav@email.com', '9876543210', 'P1234567', 'Indian'),
('Priya Mehta', 'priya@email.com', '9123456789', 'P2345678', 'Indian'),
('John Smith', 'john@email.com', '+447911123456', 'GB9876543', 'British'),
('Fatima Al-Hassan', 'fatima@email.com', '+971501234567', 'AE3456789', 'Emirati'),
('Chen Wei', 'chenwei@email.com', '+8613012345678', 'CN4567890', 'Chinese');
```

✈ Insert Bookings & Payments

```
INSERT INTO Bookings (passenger_id, seat_id, flight_id, booking_status) VALUES
(1, 1, 1, 'Confirmed'),
(2, 5, 1, 'Confirmed'),
(3, 9, 2, 'Waitlisted'),
(4, 13, 2, 'Confirmed'),
(5, 17, 3, 'Cancelled');

INSERT INTO Payments (booking_id, amount, payment_method, payment_status) VALUES
(1, 8500.00, 'CreditCard', 'Paid'),
(2, 8500.00, 'UPI', 'Paid'),
(3, 21250.00, 'NetBanking', 'Pending'),
(4, 34000.00, 'CreditCard', 'Paid'),
(5, 0.00, 'Wallet', 'Refunded');
```

✈ Insert Loyalty Programs

```
INSERT INTO Loyalty_Programs (passenger_id, tier, points_balance) VALUES
(1, 'Gold', 15000),
(2, 'Silver', 7500),
(3, 'Platinum', 42000),
(4, 'Bronze', 1200),
(5, 'Gold', 18500);
```

✈ QUERIES

✈ View All Scheduled Flights

```
SELECT f.flight_id, f.flight_number, al.airline_name,
       a1.city AS origin, a2.city AS destination,
       f.departure_time, f.arrival_time, f.status
FROM Flights f
JOIN Airlines al ON f.airline_id = al.airline_id
JOIN Routes r    ON f.route_id = r.route_id
JOIN Airports a1 ON r.origin_airport_id = a1.airport_id
JOIN Airports a2 ON r.destination_airport_id = a2.airport_id
WHERE f.status = 'Scheduled';
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

flight_id	flight_number	airline_name	origin	destination	departure_time	arrival_time	status
1	AI101	Air India	New Delhi	Mumbai	2025-06-10 06:00:00	2025-06-10 08:15:00	Scheduled
2	EK501	Emirates	New Delhi	Dubai	2025-06-10 10:30:00	2025-06-10 13:00:00	Scheduled
3	BA117	British Airways	Mumbai	London	2025-06-11 22:00:00	2025-06-12 04:30:00	Scheduled

✈ Check Available Seats on a Flight

```
SELECT s.seat_number, fc.class_name, s.status
FROM Seats s
JOIN Fare_Classes fc ON s.fare_class_id = fc.fare_class_id
WHERE s.flight_id = 1 AND s.status = 'Available'
ORDER BY fc.class_name, s.seat_number;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

seat_number	class_name	status
14D	Economy	Available
31C	Economy	Available
32A	Economy	Available
12A	Business	Available
12B	Business	Available
2A	First	Available

✈ Passenger Booking Status

```
SELECT b.booking_id, p.full_name, f.flight_number,
       s.seat_number, fc.class_name, b.booking_status, b.booking_date
FROM Bookings b
JOIN Passengers p   ON b.passenger_id = p.passenger_id
JOIN Flights f      ON b.flight_id = f.flight_id
JOIN Seats s        ON b.seat_id = s.seat_id
JOIN Fare_Classes fc ON s.fare_class_id = fc.fare_class_id;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

booking_id	full_name	flight_number	seat_number	class_name	booking_status	booking_date
1	Aarav Sharma	AI101	14C	Economy	Confirmed	2026-04-05 20:34:46
2	Priya Mehta	AI101	15A	Economy	Confirmed	2026-04-05 20:34:46
5	Chen Wei	BA117	22D	Economy	Cancelled	2026-04-05 20:34:46
3	John Smith	EK501	4B	Business	Waitlisted	2026-04-05 20:34:46
4	Fatima Al-Hassan	EK501	1A	First	Confirmed	2026-04-05 20:34:46

✈ Payment Records

```
SELECT p.full_name, f.flight_number, py.amount,
       py.payment_method, py.payment_status, py.payment_date
FROM Payments py
JOIN Bookings b  ON py.booking_id = b.booking_id
JOIN Passengers p ON b.passenger_id = p.passenger_id
JOIN Flights f    ON b.flight_id = f.flight_id;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	flight_number	amount	payment_method	payment_status	payment_date
Aarav Sharma	AI101	8500.00	CreditCard	Paid	2026-04-05 20:34:46
Priya Mehta	AI101	8500.00	UPI	Paid	2026-04-05 20:34:46
Chen Wei	BA117	0.00	Wallet	Refunded	2026-04-05 20:34:46
John Smith	EK501	21250.00	NetBanking	Pending	2026-04-05 20:34:46
Fatima Al-Hassan	EK501	34000.00	CreditCard	Paid	2026-04-05 20:34:46

✈ Overbooking — Waitlisted Passengers

```
SELECT p.full_name, f.flight_number, b.booking_id,
       lp.tier AS loyalty_tier, lp.points_balance
FROM Bookings b
JOIN Passengers p     ON b.passenger_id = p.passenger_id
JOIN Flights f        ON b.flight_id = f.flight_id
LEFT JOIN Loyalty_Programs lp ON p.passenger_id = lp.passenger_id
WHERE b.booking_status = 'Waitlisted'
ORDER BY lp.tier DESC, lp.points_balance DESC;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	flight_number	booking_id	loyalty_tier	points_balance
John Smith	EK501	3	Platinum	42000

✈ Cancellations and Refund Status

```
SELECT p.full_name, f.flight_number, c.reason,
       c.refund_status, c.cancelled_at
FROM Cancellations c
JOIN Bookings b   ON c.booking_id = b.booking_id
JOIN Passengers p ON b.passenger_id = p.passenger_id
JOIN Flights f    ON b.flight_id = f.flight_id;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	flight_number	reason	refund_status	cancelled_at
Chen Wei	BA117	Personal emergency	Processed	2025-06-09 14:22:00

✈ Total Revenue per Flight

```
SELECT f.flight_number, al.airline_name,
       COUNT(b.booking_id)   AS total_bookings,
       SUM(py.amount)        AS total_revenue
FROM Flights f
JOIN Airlines al ON f.airline_id = al.airline_id
JOIN Bookings b  ON f.flight_id = b.flight_id
JOIN Payments py ON b.booking_id = py.booking_id
WHERE py.payment_status = 'Paid'
GROUP BY f.flight_number, al.airline_name
ORDER BY total_revenue DESC;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

flight_number	airline_name	total_bookings	total_revenue
EK501	Emirates	1	34000.00
AI101	Air India	2	17000.00

✈ Flight Load Factor (Seat Occupancy %)

```
SELECT f.flight_number,
       f.total_seats,
       COUNT(CASE WHEN s.status = 'Booked' THEN 1 END) AS booked_seats,
       ROUND(COUNT(CASE WHEN s.status = 'Booked' THEN 1 END)
             * 100.0 / f.total_seats, 2) AS load_factor_pct
FROM Flights f
LEFT JOIN Seats s ON f.flight_id = s.flight_id
GROUP BY f.flight_id, f.flight_number, f.total_seats;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

flight_number	total_seats	booked_seats	available_seats	load_factor_pct
AI101	180	2	6	1.11
EK501	220	2	6	0.91
BA117	200	1	3	0.50

✈ Pending Payments

```
SELECT p.full_name, f.flight_number,
       py.amount, py.payment_method, py.payment_date
FROM Payments py
JOIN Bookings b   ON py.booking_id = b.booking_id
JOIN Passengers p ON b.passenger_id = p.passenger_id
JOIN Flights f    ON b.flight_id = f.flight_id
WHERE py.payment_status = 'Pending';
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	flight_number	amount	payment_method	payment_date
John Smith	EK501	21250.00	NetBanking	2026-04-05 20:34:46

✈ Full Reservation Report

```
SELECT p.full_name, p.passport_no, f.flight_number,
       a1.city AS from_city, a2.city AS to_city,
       s.seat_number, fc.class_name,
       b.booking_status, py.amount, py.payment_status,
       lp.tier AS loyalty_tier
FROM Bookings b
JOIN Passengers p       ON b.passenger_id = p.passenger_id
JOIN Flights f          ON b.flight_id = f.flight_id
JOIN Routes r           ON f.route_id = r.route_id
JOIN Airports a1        ON r.origin_airport_id = a1.airport_id
JOIN Airports a2        ON r.destination_airport_id = a2.airport_id
JOIN Seats s            ON b.seat_id = s.seat_id
JOIN Fare_Classes fc    ON s.fare_class_id = fc.fare_class_id
LEFT JOIN Payments py   ON b.booking_id = py.booking_id
LEFT JOIN Loyalty_Programs lp ON p.passenger_id = lp.passenger_id;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	passport_no	flight_number	from_city	to_city	seat_number	class_name	booking_status	amount	payment_status	loyalty_tier
Aarav Sharma	P1234567	AI101	New Delhi	Mumbai	14C	Economy	Confirmed	8500.00	Paid	Gold
Priya Mehta	P2345678	AI101	New Delhi	Mumbai	15A	Economy	Confirmed	8500.00	Paid	Silver
John Smith	GB9876543	EK501	New Delhi	Dubai	4B	Business	Waitlisted	21250.00	Pending	Platinum
Fatima Al-Hassan	AE3456789	EK501	New Delhi	Dubai	1A	First	Confirmed	34000.00	Paid	Bronze
Chen Wei	CN4567890	BA117	Mumbai	London	22D	Economy	Cancelled	0.00	Refunded	Gold